Lucas Lachaud

Toulouse, France | lucas.lachaud@etu.toulouse-inp.fr
LinkedIn: linkedin.com/in/lucas-lachaud | Portfolio: lucas-lachaud.github.io/Sublime-ePortfolio
French citizen (EU) | Eligible for a Singapore Work Holiday Pass | Available for a summer 2027 internship

Education

Toulouse INP – ENSEEIHT Toulouse, France
Master's-level Engineering Degree in Computer Science – Architecture, Systems & Networks *Expected 2028*
Admitted directly into the second year of the engineering programme.

University of New Caledonia Nouméa, New Caledonia
BSc in Computer Science – Highest Honours (Mention Très Bien), Ranked 1st in cohort *Feb 2024 – Jun 2026*
Relevant Coursework: Software Engineering, Java OOP, Algorithms, Databases, Operating Systems, Networks, Cryptography

Experience

Banque Calédonienne d'Investissement (BCI) Nouméa, New Caledonia
Software Engineer Intern – Digital Banking Platform *Oct 2025 – Jan 2026*
New Caledonia's leading bank, part of Groupe BRED

- Delivered 28 user stories for BCInet, the bank's online banking platform (80,000+ subscribers, ~30,000 daily visits), in an Agile Scrum team with 3-week sprints and monthly releases.
- Built internal back-office features end to end (Angular, Spring Boot REST APIs, JPA): redesigned transfers view, new direct-debits module and multi-criteria client search.
- Developed a daily Spring Batch job that automatically ends temporary subscription discounts and updates billing in the core banking system.
- Shipped customer-facing web and mobile features on a proprietary e-banking platform (Java, SQL), such as standing-order totals and savings-account cap warnings.
- Wrote JUnit tests and reviewed pull requests; CI/CD and code-quality gates with Azure DevOps and SonarQube.

Projects

Secure RSA Encryption Library 2025
Tutored project, team of two | C, GMP, libsodium | Code on GitHub

- Built a 4096-bit RSA library in C: key generation with Miller–Rabin, GMP for big-number arithmetic, libsodium for secure randomness.
- Implemented and benchmarked five modular-exponentiation algorithms, including side-channel-resistant ladders from a 2021 research paper: the most resistant is 3.6× slower than square-and-multiply (0.47 s vs 0.13 s per decryption).

Home Server 2026
Personal project | Linux, Tailscale, 3D printing

- Rebuilt an AMD BC-250 board (originally designed for crypto mining) into a Linux home server with secure remote access via Tailscale, in a case I designed and 3D-printed.

Skills & Interests

Programming: Java, C, TypeScript, JavaScript, Python, SQL

Frameworks & Backend: Spring Boot, Spring Batch, JPA/Hibernate, Angular, JUnit, Maven

Systems & DevOps: Linux, Docker, Git, Azure DevOps, CI/CD, SonarQube, Liquibase, Postman, DBeaver

Security: Cryptography, side-channel countermeasures

Languages: French (Native), English (B2), Spanish (A2)

Interests: Financial markets, astrophotography, aviation (started private pilot training), electronics and 3D printing